

Galgotias University, Greater Noida, U.P.

School of Computer Science & Engineering

Department of Artificial Intelligence & Data Science

Web Technology (R1UC626C)

Project Synopsis

Title : Teacher Availability System	
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Introduction:

In educational institutions, students often face difficulty in locating teachers or knowing their availability for doubt-solving or academic discussions. This leads to wasted time and inefficient communication. The Teacher Availability System is a web-based solution designed to bridge this gap by providing real-time teacher availability, appointment booking, and location tracking within the campus.

Objective:

- To provide real-time availability status of teachers
- To enable students to book appointments easily
- To reduce waiting time and improve communication
- To digitize teacher-student interaction within campus

Technology Stack:

- Frontend: HTML, CSS, JavaScript / React
- Backend: Node.js / Spring Boot
- Database: MySQL / MongoDB
- Real-time Communication: WebSockets
- Authentication: JWT / Session-based
- Deployment: Cloud (AWS / Firebase)

Methodology:

The system provides a smart dashboard for both teachers and students:

For Teachers:

- Set availability slots (time-based schedule)
- Update real-time status:
 - Available
 - Busy
 - Offline
- Update current location:
 - In Cabin
 - In Class
 - Not Available

For Students:

- View teacher availability in real-time
- Book appointments based on available slots
- Receive notifications for approval/rejection

Features:

- **Dynamic Product Browsing:** Fully functional search bar, category filtering, and detailed product view pages.
- **Database Integration:** Fetching and displaying "Recommended for You" and "Best Sellers" dynamically from a backend server rather than hardcoded HTML.

•**State Management:** Persistent shopping cart that retains items even as the user navigates between different views or refreshes the page.

•**Advanced Custom Functionalities:** Implementation of user authentication (login/signup), a simulated checkout and payment gateway process, and an admin dashboard for dynamic inventory management.

•**User Interactions:** Hover states, active UI feedback, and responsive design for both desktop and mobile viewports.

Applications:

- Improved teacher-student interaction
- Reduced time wastage
- Better scheduling and productivity
- Smart digital campus experience

Conclusion:

The Teacher Availability System is a practical and scalable solution for modern educational institutions. By combining real-time technology with smart scheduling, it enhances communication efficiency and improves overall academic productivity.